
CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Kalyanapu, Alfred

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0002-7746-3121>

*POSITION TITLE: Professor

*ORGANIZATION AND LOCATION: Tennessee Tech University, Cookeville, Tennessee, United States

Proposals and Active Projects

*Proposal/Active Project Title: Modeling and Simulation of Cascading Dam Failures using HEC-RAS

*Status of Support: Current

Proposal/Award Number:

*Source of Support: Tennessee Department of Environment and Conservation

*Primary Place of Performance: Tennessee Tech University

*Proposal/Active Project Start Date: (MM/YYYY): 11/2024

*Proposal/Active Project End Date: (MM/YYYY): 04/2026

*Total Anticipated Proposal/Project Amount: \$100,053

* Person Months per budget period Devoted to the Proposal/Active Project:

Year	Person Months
2025	1

*Overall Objectives:

The overall objective of this project is to develop a Hydrologic Engineering Center River Analysis System (HEC-RAS) model for a series of two dams in Scott County, Tennessee, and simulate dam breach by cascading failure of this dam network and generate resultant flood inundation extent.

*Statement of Potential Overlap:

None

***Proposal/Active Project Title:** Development of a low-cost camera-based turbidity sensor

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** USGS (104b)

***Primary Place of Performance:** Tennessee Tech University

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2024

***Proposal/Active Project End Date: (MM/YYYY):** 08/2025

***Total Anticipated Proposal/Project Amount:** \$34,595

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.61

***Overall Objectives:**

The proposed project will aim to develop a proof-of-concept low-cost camera-based turbidity sensor for surface waterbodies using secci disks. The design of the sensor will focus on making it easily deployable, and reliable compared to commercially available turbidity sensors.

***Statement of Potential Overlap:**

N/A

***Proposal/Active Project Title:** Identifying tradeoffs in ecosystem services in restored agricultural wetlands in the Agricultural Conservation Easements Program-Wetland Reserve Easements (ACEP-WRE) program.

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** USDA NRCS

***Primary Place of Performance:** Tennessee Tech University

***Proposal/Active Project Start Date: (MM/YYYY):** 10/2023

***Proposal/Active Project End Date: (MM/YYYY):** 09/2026

***Total Anticipated Proposal/Project Amount:** \$299,961

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2024	0.5
2025	0.5
2026	0.5

***Overall Objectives:**

Project Objectives:

- Determine the relationship between nutrient (nitrogen and phosphorus) retention, and GHG (CO₂, N₂O, CH₄) production for major WRP/WRE restoration practices.
- Determine the environmental conditions and mechanisms that regulate nutrient retention and GHG production in both surface and shallow groundwater environments.
- Define the relationship between MESA and nitrate concentrations in surface waters of western Tennessee and Kentucky WRP/WRE easements.

***Statement of Potential Overlap:**

N/A

***Proposal/Active Project Title:** AI-Driven Observable Synthesis for the INQ Electronic Structure Engine via Large Language Models: Structure Engine via Large Language Models: Accelerated Materials Discovery for the Genesis Mission Accelerated Materials Discovery for the Genesis Mission

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** DOE

***Primary Place of Performance:** Tennessee Tech University

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 03/2027

***Total Anticipated Proposal/Project Amount:** \$230,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.5

***Overall Objectives:**

This Phase I Genesis project responds to Topic 18-B with a transformative approach to enhancing

DOE-relevant applications. Investigators at TN TECH, LLNL, and UT Chattanooga will explore revolutionary approaches to using AI to generate new observables in an important LLNL Quantum Materials Code (INQ). The results of this work will be generalizable across the DOE complex of large-scale applications.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:** Optimizing Floodplain Nutrient Retention with AI Controlled Gates

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** EPA

***Primary Place of Performance:** Tennessee Tech University

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2026

***Proposal/Active Project End Date: (MM/YYYY):** 12/2028

***Total Anticipated Proposal/Project Amount:** \$993,953

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1
2027	1
2001	1

***Overall Objectives:**

This demonstration project integrates ecology, environmental engineering, and computer science disciplines to create a machine learning-based autonomous water control gate that optimizes floodplain nutrient retention by deciding how much and how long to keep water in a riparian area. The initial control model is based on previous regional wetland nutrient retention research. However, this control system will continuously self-improve its performance through constant real-time data input from local sensors and remote data including hydrology and nutrient concentrations. Through field-scale testing and validation, this project will create a more efficient and cost-effective tool for water resource managers to improve water quality.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:** Development of a Machine learning based low-cost camera-based turbidity sensor

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** USGS

***Primary Place of Performance:** Tennessee Tech University

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2025

***Proposal/Active Project End Date: (MM/YYYY):** 08/2026

***Total Anticipated Proposal/Project Amount:** \$35,206

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.01

***Overall Objectives:**

The proposed project is the second phase of our research focused on developing a proof-of-concept low-cost camera-based turbidity sensor for surface waterbodies using secchi disks. Additionally, an alternative design of the camera-based sensor will be explored using low-cost analog sensors that use nephelometric mode for measuring the turbidity. The design of the sensor will focus on making it easily deployable, and reliable compared to commercially available turbidity sensors. The prototype developed in this project will be tested under laboratory and field conditions. Streams of various sizes within middle and eastern Tennessee will be selected as test sites. Additionally, the project team will leverage existing collaboration with researchers in India where the functionality of the sensors will be tested in selected sites to evaluate their applicability to a wide range of site conditions.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:** Watershed-wide Stormwater Management for the Town of Gainesboro

***Status of Support:** Past

Proposal/Award Number:

***Source of Support:** Tennessee Department of Environment and Conservation

***Primary Place of Performance:** Cookeville

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2024

***Proposal/Active Project End Date: (MM/YYYY):** 08/2025

***Total Anticipated Proposal/Project Amount:** \$101,047

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2024	0.2
2025	0.15

***Overall Objectives:**

- Objective 1: Develop select criteria of water features to be assessed for water quality and water levels at pre-storm, storm and post-storm events.
- Objective 2: Monitor water quality, water level and relevant weather data in the Doe Creek watershed for a continuous 12-month period.
- Objective 3: Using previously developed hydraulic model, water level, weather and water quality data, as well as stormwater infrastructure survey data to be provided by the Town of Gainesboro, develop a Stormwater Management Model (SWMM) to evaluate stormwater best management alternatives for the Doe Creek watershed. Note, this objective can only be accomplished once all stormwater asset survey is completed by the Town of Gainesboro.

***Statement of Potential Overlap:**

N/A

***Proposal/Active Project Title:** CC* Compute: A HPC Cluster for Science Research and Education at Tennessee Tech University

***Status of Support:** Past

Proposal/Award Number: 30876

***Source of Support:** NSF

***Primary Place of Performance:** Tennessee Technological University

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2021

***Proposal/Active Project End Date: (MM/YYYY):** 08/2024

***Total Anticipated Proposal/Project Amount:** \$399,983

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2021	0.01

***Overall Objectives:**

Addition of state-of-the-art, cost-effective 10 node GPU cluster supporting 20 NVIDIA A100 GPUs and 1280 AMD Epyc2 CPU cores, to the existing shared campus high-performance computing infrastructure.

***Statement of Potential Overlap:**

Doesn't overlap with any active or pending projects.

***Proposal/Active Project Title:** Developing a Watershed Plan for the Falling Water River Watershed

***Status of Support:** Past

Proposal/Award Number:

***Source of Support:** Tennessee Department of Environment and Conservation

***Primary Place of Performance:** Tennessee Tech University

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2021

***Proposal/Active Project End Date: (MM/YYYY):** 06/2024

***Total Anticipated Proposal/Project Amount:** \$73,041

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2024	0.25

***Overall Objectives:**

Initiate development of a management and protection plan for the Falling Water River watershed in Tennessee.

***Statement of Potential Overlap:**

N/A

Addendum

Do the Notice of Funding Opportunity (NOFO) or Award instructions require a Current and Pending (Other) Support addendum?

No

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Current and Pending (Other) Support Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

If this is an R&D proposal/award, I also certify to the following statement:

Within the past 12 months I have completed research security training meeting the requirements in SEC. 10634(b) of 42 USC 19234.

Certified by Kalyanapu, Alfred in SciENCv on 2026-04-19 21:59:16